

Project Documentation

Title: *Sales Analysis Dashboard using Microsoft Excel*

1. Problem Statement

In many organizations, sales data is stored in large tables, which makes it difficult to analyze manually. It becomes hard to understand sales trends, region-wise sales, category performance, and monthly growth without proper visualization.

The objective of this project is to create an **interactive Sales Analysis Dashboard in Microsoft Excel** that allows users to easily analyze sales data using filters, charts, and pivot tables.

The dashboard helps in:

- Analyzing sales by region
- Analyzing sales by category
- Analyzing sales by product
- Analyzing monthly sales trend
- Comparing sales with target
- Filtering data dynamically

This project solves the problem by creating a **dynamic and user-friendly dashboard** using Excel tools.

2. Objective of the Project

- To design a Sales Analysis Dashboard in Excel
 - To summarize large sales data using Pivot Tables
 - To visualize data using charts
 - To use slicers for filtering
 - To calculate target hit percentage
 - To create a professional dashboard layout
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3. Data Description

The dataset contains sales records with the following columns:

Column Name	Description
Date	Date of sale
Month	Month extracted from date
Region	Sales region
Category	Product category
Product	Product name
Sales	Sales amount
Quantity	Quantity sold
Target	Fixed target value (400)
Target Hit %	Sales / Target
Away from Target %	Difference from target

5. Tools Used

Tool	Purpose
Microsoft Excel	Creating dashboard
Pivot Table	Data summary

Pivot Chart	Visualization
Slicer	Filtering
Excel Formulas	Calculations
VBA Macro	Reset filters, Summary Report
Shapes & Text Box	KPI design

6. Features Used in Project

- Pivot Tables added for data summarization
 - Pivot Charts added for visualization
 - Slicers added for filtering the dashboard
 - KPI Cards added for showing key values
 - Macro added for resetting filters and Summary Report
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7. Dashboard Details

The Sales Analysis Dashboard contains:

The Sales Analysis Dashboard is created using Microsoft Excel to analyze sales data using Pivot Tables, Charts, KPIs, and Slicers.

The dashboard is interactive and updates automatically when filters are applied.

The dashboard contains the following components:

- Slicers (Filters)
- KPI Cards
- Pivot Tables
- Pivot Charts
- Reset Filter Button

Each component is explained below.

1. Slicers (Filters)

Slicers are used to filter the entire dashboard dynamically.

Region Slicer

Used to filter data by region.

Values:

- East
- North
- South
- West

Use:

Shows sales only for selected region.

Category Slicer

Values:

- Beverages
- Dairy
- Snacks

Use:

Filters dashboard based on category.

Salesperson Slicer

Values:

- Amit
- Neha
- Priya
- Raj

Use:

Shows data for selected salesperson.

2. KPI Cards

These show important values.

KPI 1 — Total Sales

Shows total of all sales.

Used for:

Overall sales summary.

Formula:

Sum of Sales column.

KPI 2 — Target Achievement %

Shows how much target is achieved.

Formula:

Target Hit % = Sales / Target

Used for:

Checking if sales reached target.

KPI 3 — Top Region

Shows region with highest sales.

Used for:

Finding best performing region.

Pivot used:

Region vs Total Sales

KPI 4 — Top Category

Shows category with highest sales.

Used for:

Finding best category.

Pivot used:

Category vs Sales

KPI 5 — Top Product

Shows product with highest sales.

Pivot used:

Product vs Sales

Use:

Find best selling product.

KPI 6 — Top Salesperson

Shows salesperson with highest sales.

Pivot used:

Salesperson vs Sales

Use:
Find best performer.

3. Pivot Table

Pivot Table 1 — Sales by Salesperson

Columns:

- Salesperson
- Total Sales

Use:

Compare sales of each person.

Used for:

Pie chart

KPI

Bar chart

Pivot Table 2 — Target Hit %

Columns:

- Salesperson
- Target Hit %

Use:

Check who achieved target.

Used for:

Target chart

Pivot Table 3 — Away from Target %

Columns:

- Salesperson
- Away from Target %

Use:

Check how far from target.

Negative → below target

Positive → above target

Pivot Table 4 — Region Total Sales

Columns:

- Region
- Total Sales

Use:

Find region performance.

Used in:

Region chart

Top region KPI

Pivot Table 5 — Month Total Sales

Columns:

- Month
- Total Sales

Use:

Monthly trend.

Used in:

Line chart.

Pivot Table 6 — Category Total Sales

Columns:

- Category
- Quantity
- Sales

Use:

Category comparison.

Used in:

Category chart.

Pivot Table 7 — Product Total Sales

Columns:

- Product
- Total Sales

Use:

Product analysis.

Used in:

Product chart.

4. Charts

Charts — Sales by Month (Line Chart)

Use:

Shows trend over time.

Purpose:

Check increase / decrease.

Used pivot:

Month vs Sales.

Chart — Sales by Salesperson (Pie Chart)

Use:

Shows percentage share.

Purpose:

Compare contribution.

Used pivot:

Salesperson vs Sales.

Chart — Sales by Target Hit % (Clustered Bar)

Use:

Shows target achievement.

Purpose:

Compare performance.

Used pivot:

Salesperson vs Target %.

Chart — Sales by Region (Stacked Column)

Use:

Compare regions.

Purpose:

Find best region.

Used pivot:
Region vs Sales.

Chart — Sales by Category and Region (Stacked Bar)

Use:
Compare category in each region.
Purpose:
Detailed analysis.
Used pivot:
Category + Region.

Chart — Sales by Product (Clustered Column)

Use:
Compare products.
Purpose:
Find best product.
Used pivot:
Product vs Sales.

Chart — Category by Sales and Quantity (Clustered Column)

Use:
Compare quantity and sales.
Purpose:
Check demand vs revenue.
Used pivot:
Category vs Qty vs Sales.

5. Macros

Reset Filters Button

Macro used: Clear all slicers and filters.
Use:
Clear all slicers.
Purpose:
Return dashboard filters & slicers to default.

Summary Report Macro

A macro was created to generate a summary report automatically.
This macro copies KPI values from the dashboard and creates a formatted summary in a new sheet named **Summary**.
It also displays overall sales result and region-wise analysis based on the dashboard.
Use:
Generate automatic summary report.
Purpose:
Reduce manual work and create quick report.

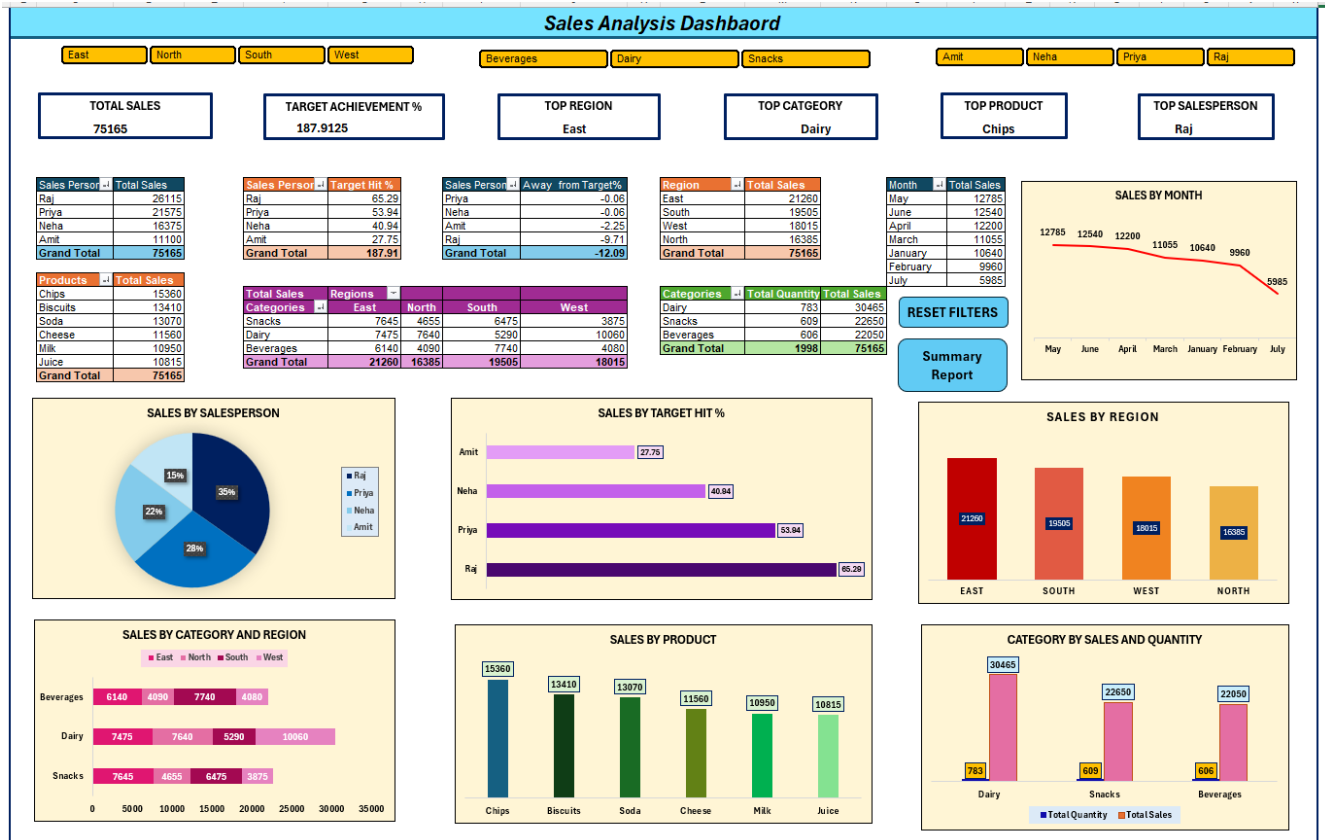
6. Advantages

- Easy to understand
- Fast analysis
- Interactive dashboard
- No manual calculation
- Professional report

7. Limitations

- Data must be updated manually
- Works only in Excel
- Large data may slow file

8. Results



SALES SUMMARY REPORT			
Total Sales	75165		
Target Achievement %	187.9125		
Top Region	East		
Top Category	Dairy		
Top Product	Chips		
Top Salesperson	Raj		
SUMMARY			
Total sales is 75165.			
Dairy is the best category selling.			
East region has highest sales.			
Raj is top performer overall with highest sales.			
North region needs improvement.			
Region Wise Analysis			
East - Highest category Snacks, Highest product Chips, Best performer Raj			
North - Highest category Dairy, Highest product Cheese, Best performer Neha			
South - Highest category Beverages, Highest product Chips, Best performer Raj			
West - Highest category Dairy, Highest product Cheese, Best performer Raj			

9. Conclusion

The Sales Analysis Dashboard was created using Microsoft Excel with Pivot Tables, Charts, Slicers, and KPI cards. The dashboard helps in easy and quick analysis of sales data using interactive filters and visual charts.

This project demonstrates the use of Excel for data analysis and dashboard development.